

Tarun Konagalla

tarunkonagalla2806@gmail.com | [LinkedIn](#) | Austin, TX | +1-945-274-8351 | [Portfolio](#)

Work Experience

Software Development Engineer

Austin, TX

Amazon Web Services

(Sep 2025 – Present)

- Engineer and operate Java-based, event-driven systems for AWS IAM Identity Center using Lambda, SQS, SNS, and DynamoDB, processing lifecycle events across 30+ AWS Regions.
- Implemented ARN-based authorization controls for multi-Region identity replication, tightening access enforcement for replicated resources and strengthening cross-Region resiliency.
- Automated deployments to new AWS Regions using configuration-driven infrastructure and CI/CD approval workflows, cutting manual steps and improving deployment consistency and safety.
- Improved event-processing reliability by implementing retry logic, failure handling, dead-letter queue recovery, and Region-aware routing for workflows spanning 30+ Regions.
- Built global CloudWatch dashboards and alarms covering queue backlogs, processing failures, failed events, dead-letter queue growth, and regional service degradation, enabling faster detection and troubleshooting.
- Support production services through a 24/7 on-call rotation, diagnosing system failures, performing root-cause analysis, recovering failed events, and coordinating cross-team fixes.

Volunteer Software Engineer

Remote

Saayam For All

(Jun 2025 – Oct 2025)

- Developed full-stack features using React, Java, and Spring Boot, delivering 3 reusable UI components and 10+ REST APIs for end-to-end workflows.
- Integrated Spring Boot services with PostgreSQL and Amazon Aurora, supporting reliable data access for production application features.
- Collaborated with a remote team, translating requirements into technical changes and delivering 7 planned application features through code reviews and documentation.
- Diagnosed and resolved 6+ bugs in production, ensuring system stability and minimizing user impact.

Web Technology Assistant

Dallas, TX

Southern Methodist University

(Aug 2024 – Jun 2025)

- Maintained, tested, and delivered functionality and content updates across 30+ university web pages using Sitecore CMS and Azure SQL, ensuring accurate and consistent production releases.
- Partnered with 10+ academic and administrative departments to translate publishing requirements into technical updates and resolve accessibility, content, functionality, and layout issues.
- Troubleshoot Sitecore publishing failures, broken page configurations, database-related discrepancies, and environment-specific issues across development and production systems.
- Led Agile planning, prioritised incoming technical requests, performed pre-production validation, and coordinated time-sensitive website updates with stakeholders.
- Documented publishing workflows, recurring issue patterns, and resolution procedures, improving support consistency and reducing repeated troubleshooting effort across departments.

Projects

ChangeLens – Document Change Intelligence Platform

Java, Spring Boot, React, PostgreSQL

- Built a full-stack document-comparison platform that identifies changes to dates, costs, responsibilities, and requirements across document versions.
- Designed a section-alignment and change-classification pipeline combining deterministic text comparison, structured extraction, and user confirmation.
- Implemented secure document storage, comparison history, exportable reports, and unit and integration tests using JUnit, Mockito, and Testcontainers.

Abnormal Activity Detection in Surveillance Footage

Published in IJIEMR, Apr 2023

- Developed abnormal activity detection pipeline using Python, OpenCV, and PyTorch, achieving a 99.991% testing score with VGG-16.
- Preprocessed 6,047 extracted frame samples by standardizing video frames and resizing inputs to $150 \times 150 \times 3$ for consistent model training.

Education

Southern Methodist University, MS in Software Engineering (GPA - 3.7)

Dallas, Texas

(Aug 2023 – May 2025)

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL

Backend: Spring Boot, Node.js, REST APIs, Microservices

Cloud & Infrastructure: Lambda, SQS, SNS, DynamoDB, CloudWatch, IAM, API Gateway, EC2, S3, RDS, EKS, VPC, Route 53, CloudFormation, CDK, SAM

Databases: PostgreSQL, Amazon Aurora, MySQL, MongoDB

Testing & Tools: Unit Testing, Integration Testing, JUnit, Mockito, Git, Docker, Jenkins, CI/CD, Linux

Core Concepts: Distributed Systems, Event-Driven Architecture, Infrastructure as Code, Serverless Computing, System Reliability